EPSR-test-ex62 chains: H-bond chains, multi-criterion distribution

Program(s): chains | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `chains` [1], which finds chains of bonded molecules with the shortest-path criterion of Franzblau [2] as cited in the Fortran: a bond requires an atom1-atom2 pair in one molecule and atom3 in another within three distance windows (and optionally an atom2-atom1-atom3 angle range); chains start at molecules with one neighbour, and histograms of neighbour numbers and chain lengths are written.

Code exercised

Fortran (original, fortran_src/):

- `chains_sp_criterion.f`: `run_chains`, `find_chains` (bond search, chain search, histograms)
- `chains_routines.f`: `match` (wildcard `*`), `shortest_path`
- `read_and_save_files.f`: `outgofz`, `save_files_others`; `setup_symbols_generic.f`: `setup_symbols_chains` (computed `nsize` default)

Rust (port, rust/epsr-core/src/):

- `tools/chains.rs`: `run_chains_or_rings`, `read_criteria`, `find_bonds`, `find_chains`, `shortest_path`, `fmatch`
- `tools/clusters.rs`: `outgofz`, `open_input_with`; `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `CHAINS`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å³), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [3]. Distribution 1: water-water hydrogen bonds O1(-H1)···O1 with O-H 0.8-1.2 Å, O···O 2.5-3.3 Å, H···O 1.5-2.4 Å and H-O···O angle 0-30°. Distribution 2: two criteria (`iw` loop), `O1 H1 O*` (`*` matches DMSO O and water O1, angle 0-35°) and `O1 H1 S` (0-180°, angle test skipped because the cosine range is >= 1.9). Reference `fort.14`: mean neighbour numbers 1.37 and 2.25.

Inputs

`hbchain.CHAINS.dat`: `nsize 40`, `nsumt 0`, two distributions as above. `200DMSO400Water.ato` copied unchanged from EPSR-test; the keyword file was written by hand (the Fortran program creates no template when the named file does not exist; keywords and defaults from `setup_symbols_generic.f`).

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2` with `nsumt 0` (accumulation over the two runs; reference `nsumt 2`). Exercises the bond search, `shortest_path`, the chain search and both histograms. Compared after each pass: `hbchain.CHAINS.n01`, the rewritten `hbchain.CHAINS.dat`, and `fort.14` (mean and spread of each histogram, written after unit 14 was closed, see ex60). The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] D. S. Franzblau, "Computation of ring statistics for network models of solids", Phys. Rev. B 44, 4925-4930 (1991).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.